

Cooling System Draining, Filling and Bleeding

 [Ford Ecat Electronic Parts Catalog](#)

Special Tool(s)

	Air Lift Cooling System Tester UUV550000 or equivalent
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Material

Item	Specification
Motorcraft® Orange Antifreeze/Coolant Concentrated VC-3-B (US); CVC-3-B2 (Canada)	WSS-M97B44-D
Motorcraft® Orange Antifreeze/Coolant Prediluted VC-3DIL-B (US); CVC-3DIL-B (Canada)	WSS-M97B44-D2

Draining

WARNING: Always allow the engine to cool before opening the cooling system. Do not unscrew the coolant pressure relief cap when the engine is operating or the cooling system is hot. The cooling system is under pressure; steam and hot liquid can come out forcefully when the cap is loosened slightly. Failure to follow these instructions may result in serious personal injury.

NOTICE: The coolant must be recovered in a suitable, clean container for reuse. If the coolant is contaminated it must be recycled or disposed of correctly. Failure to follow these instructions may result in engine or cooling system damage.

NOTE: *Less than 80% of the coolant capacity can be recovered with the engine in the vehicle. Dirty, rusty or contaminated coolant requires replacement.*

NOTE: *During normal vehicle operation, Motorcraft Orange Antifreeze/Coolant may change color from orange to pink or light red. As long as the engine coolant is clear and uncontaminated, this color change does not indicate the engine coolant has degraded nor does it require the engine coolant to be drained, the system to be flushed, or the engine coolant to be replaced.*

1. With the vehicle in NEUTRAL, position it on a hoist. For additional information, refer to Section 100-02.
2. Make sure the engine is cool.
3. Wrap a thick cloth around the pressure relief cap. Slowly turn the cap counterclockwise until the pressure begins to release.
4. Step back while the pressure releases.
5. When sure all the pressure has been released, use the cloth to turn and remove the cap.

6. Place a suitable container below the radiator draincock. Drain the coolant.
 - Tighten the radiator draincock when finished.

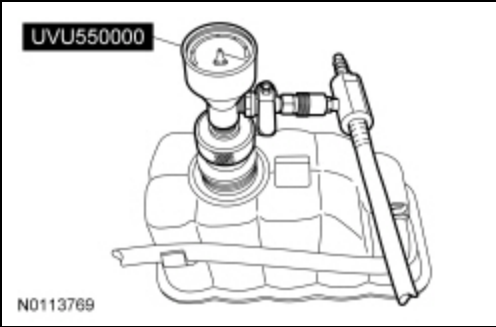
Filling and Bleeding with a Vacuum Cooling System Filler - Degas Bottle Systems

NOTICE: The engine cooling system is filled with Motorcraft® Orange Antifreeze/Coolant. Always fill the cooling system with the manufacturer's specified coolant. Chemically flush the cooling system if a non-specified coolant has been used. Refer to Cooling System Flushing in this section. Failure to follow these instructions may damage the engine or cooling system.

NOTICE: Engine coolant provides boil protection, corrosion protection, freeze protection, and cooling efficiency to the engine and cooling components. In order to obtain these protections, maintain the engine coolant at the correct concentration and fluid level in the degas bottle.

To maintain the integrity of the coolant and the cooling system:

- **Add Motorcraft® Orange Antifreeze/Coolant or equivalent. Do not mix coolant types.**
 - **Do not add or mix with any other type of engine coolant. Mixing coolants may degrade the coolant's corrosion protection.**
 - **Do not add alcohol, methanol, or brine, or any engine coolants mixed with alcohol or methanol antifreeze. These can cause engine damage from overheating or freezing.**
 - **Ford Motor Company does NOT recommend the use of recycled engine coolant in vehicles originally equipped with Motorcraft® Orange Antifreeze/Coolant since a Ford-approved recycling process is not yet available.**
1. Install the vacuum cooling system filler and follow the manufacturer's instructions to fill and bleed the cooling system. When adding or topping of the engine coolant:
 1. Measure the coolant concentration in the vehicle using the Coolant/Battery Refractometer 300-ROB75240 or equivalent.
 2. Determine the concentration desired based on the vehicle duty cycle of extreme hot or cold operating conditions.
 3. Add/top off or adjust coolant as follows:
 - For concentrations measured 48/52 to 50/50 (equates to a freeze point between -31°C [-30°F] and -34°C [-37°F]), use Motorcraft® Orange Antifreeze/Coolant Prediluted to maintain a coolant concentration in this same range.
 - For all other concentrations, use Motorcraft® Orange Antifreeze/Coolant Concentrated and/or distilled water to get to the desired concentration.
 - When refilling the engine coolant after a flush procedure, use a mixture of Motorcraft® Orange Antifreeze/Coolant Concentrated and distilled water to get to the desired concentration.
 4. Recommended coolant concentration is 48/52 to 50/50 engine coolant to distilled water (freeze protection -31°C [-30°F] to -34°C [-37°F]).
 5. For extremely cold climates (less than -37°C [-34°F]):
 - It may be necessary to increase the coolant concentration above 50%.
 - NEVER increase the coolant concentration above 60%.
 - Maximum coolant concentration is 60/40 for cold weather areas.
 - A coolant concentration of 60% provides freeze point protection down to -50°C (-58°F).
 - Engine coolant concentration above 60% decreases the overheat protection characteristics of the engine coolant and may damage the engine.
 6. For extremely hot climates:
 - It is still necessary to maintain the coolant concentration above 40%.
 - NEVER decrease the coolant concentration below 40%.
 - Minimum coolant concentration is 40/60 for warm weather areas.
 - A coolant concentration of 40% provides freeze point protection down to -26°C (-15°F).
 - Engine coolant concentration below 40% decreases the corrosion and freeze protection characteristics of the engine coolant and may damage the engine.
 7. Vehicles driven year-round in non-extreme climates should use a 48/52 to 50/50 mixture of engine coolant and distilled water (freeze protection -31°C [-30°F] to -34°C [-37°F]) for optimum cooling system and engine protection.



Filling and Bleeding without a Vacuum Cooling System Filler - Degas Bottle Systems

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To maintain the integrity of the coolant and the cooling system:

- Add Motorcraft® Orange Antifreeze/Coolant or equivalent. Do not mix coolant types.
- Do not add or mix with any other type of engine coolant. Mixing coolants may degrade the coolant's corrosion protection.
- Do not add alcohol, methanol, or brine, or any engine coolants mixed with alcohol or methanol antifreeze. These can cause engine damage from overheating or freezing.
- Ford Motor Company does NOT recommend the use of recycled engine coolant in vehicles originally equipped with Motorcraft® Orange Antifreeze/Coolant since a Ford-approved recycling process is not yet available.

1. Fill the radiator through the degas bottle until the coolant level is between the COOLANT FILL LEVEL marks. When adding or topping of the engine coolant:
 1. Measure the coolant concentration in the vehicle using the Coolant/Battery Refractometer 300-ROB75240 or equivalent.
 2. Determine the concentration desired based on the vehicle duty cycle of extreme hot or cold operating conditions.
 3. Add/top off or adjust coolant as follows:
 - For concentrations measured 48/52 to 50/50 (equates to a freeze point between -31°C [-30°F] and -34°C [-37°F]), use Motorcraft® Orange Antifreeze/Coolant Prediluted to maintain a coolant concentration in this same range.
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 - Engine coolant concentration above 60% decreases the overheat protection characteristics of the engine coolant and may damage the engine.
 6. For extremely hot climates:
 - It is still necessary to maintain the coolant concentration above 40%.
 - NEVER decrease the coolant concentration below 40%.
 - Minimum coolant concentration is 40/60 for warm weather areas.
 - A coolant concentration of 40% provides freeze point protection down to -26°C (-15°F).

- Engine coolant concentration below 40% decreases the corrosion and freeze protection characteristics of the engine coolant and may damage the engine.
7. Vehicles driven year-round in non-extreme climates should use a 48/52 to 50/50 mixture of engine coolant and distilled water (freeze protection -31°C [-30°F] to -34°C [-37°F]) for optimum cooling system and engine protection.
 2. Select the maximum heater temperature and blower motor speed settings. Position the control to discharge air at A/C vents in instrument panel.
 3. Start the engine and allow to idle. While engine is idling, feel for hot air at A/C vents.
 4. **NOTICE: If the air discharge remains cool and the Engine Coolant Temperature (ECT) gauge does not move, the engine coolant level is low and must be filled. Stop the engine, allow the engine to cool and fill cooling system. Failure to follow these instructions may result in damage to the engine.**

Start the engine and allow it to idle until normal operating temperature is reached. Hot air should discharge from A/C vents. The Engine Coolant Temperature (ECT) gauge should maintain a stabilized reading in the middle of the NORMAL range. The upper radiator hose should feel hot to the touch.
 5. Shut the engine off and allow the engine to cool.
 6. Check the engine coolant level in the degas bottle and fill as necessary.
 7. Repeat the previous 4 steps as necessary.

Filling and Bleeding with a Vacuum Cooling System Filler - Coolant Expansion Tank System

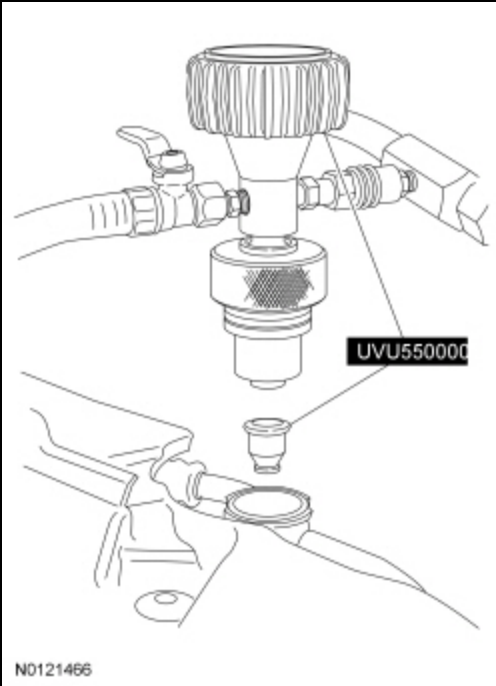
NOTICE: Vehicle cooling systems are filled with Motorcraft® Orange Antifreeze/Coolant. Always fill the cooling system with the manufacturer's specified coolant. Chemically flush the cooling system if a non-specified coolant has been used. Refer to Cooling System Flushing in this section. Failure to follow these instructions may damage the engine or cooling system.

NOTICE: Engine coolant provides freeze protection, boil protection, cooling efficiency and corrosion protection to the engine and cooling components. In order to obtain these protections, the engine coolant must be maintained at the correct concentration and fluid level. Failure to follow these instructions may damage the engine or cooling system.

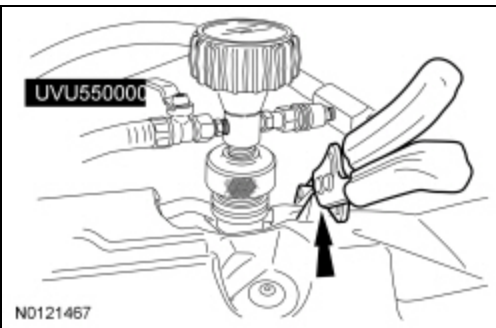
To maintain the integrity of the coolant and the cooling system:

- Add Motorcraft® Orange Antifreeze/Coolant or equivalent. Do not mix coolant types.
- Do not add or mix with any other type of engine coolant. Mixing coolants may degrade the coolant's corrosion protection.
- Do not add alcohol, methanol or brine, or any engine coolants mixed with alcohol or methanol antifreeze. These can cause engine damage from overheating or freezing.
- Ford Motor Company does NOT recommend the use of recycled engine coolant in vehicles originally equipped with Motorcraft® Orange Antifreeze/Coolant since a Ford-approved recycling process is not yet available.

1. Connect the vacuum cooling system filler and the appropriate rubber adapter to the radiator.



2. Clamp the radiator-to-coolant expansion tank hose near the radiator.



3. Follow the vacuum cooling system filler manufacturer's instructions to fill and bleed the cooling system.

When adding or topping of the engine coolant:

1. Measure the coolant concentration in the vehicle using the Coolant/Battery Refractometer 300-ROB75240 or equivalent.
2. Determine the concentration desired based on the vehicle duty cycle of extreme hot or cold operating conditions.
3. Add/top off or adjust coolant as follows:
 - For concentrations measured 48/52 to 50/50 (equates to a freeze point between -31°C [-30°F] and -34°C [-37°F]), use Motorcraft® Orange Antifreeze/Coolant Prediluted to maintain a coolant concentration in this same range.
 - For all other concentrations, use Motorcraft® Orange Antifreeze/Coolant Concentrated and/or distilled water to get to the desired concentration.
 - When refilling the engine coolant after a flush procedure, use a mixture of Motorcraft® Orange Antifreeze/Coolant Concentrated and distilled water to get to the desired concentration.
4. Recommended coolant concentration is 48/52 to 50/50 engine coolant to distilled water (freeze protection -31°C [-30°F] to -34°C [-37°F]).
5. For extremely cold climates (less than -37°C [-34°F]):
 - It may be necessary to increase the coolant concentration above 50%.
 - NEVER increase the coolant concentration above 60%.
 - Maximum coolant concentration is 60/40 for cold weather areas.
 - A coolant concentration of 60% provides freeze point protection down to -50°C (-58°F).
 - Engine coolant concentration above 60% decreases the overheat protection characteristics of the engine coolant and may damage the engine.
6. For extremely hot climates:
 - It is still necessary to maintain the coolant concentration above 40%.
 - NEVER decrease the coolant concentration below 40%.

- Minimum coolant concentration is 40/60 for warm weather areas.
 - A coolant concentration of 40% provides freeze point protection down to -26°C (-15°F).
 - Engine coolant concentration below 40% decreases the corrosion and freeze protection characteristics of the engine coolant and may damage the engine.
7. Vehicles driven year-round in non-extreme climates should use a 48/52 to 50/50 mixture of engine coolant and distilled water (freeze protection -31°C [-30°F] to -34°C [-37°F]) for optimum cooling system and engine protection.

Filling and Bleeding without a Vacuum Cooling System Filler - Coolant Expansion Tank Systems

NOTICE: Vehicle cooling systems are filled with Motorcraft® Orange Antifreeze/Coolant. Always fill the cooling system with the manufacturer's specified coolant. Chemically flush the cooling system if a non-specified coolant has been used. Refer to Cooling System Flushing in this section. Failure to follow these instructions may damage the engine or cooling system.

NOTICE: Engine coolant provides freeze protection, boil protection, cooling efficiency and corrosion protection to the engine and cooling components. In order to obtain these protections, the engine coolant must be maintained at the correct concentration and fluid level. Failure to follow these instructions may damage the engine or cooling system.

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Vehicles equipped with a 6.2L engine

1. **NOTE:** For coolant flow diagrams, refer to *Engine Cooling in Description and Operation* in this section.

Release the clamp and disconnect the heater inlet hose from the heater inlet tube.

2. Fill the radiator through the radiator fill neck until engine coolant flows from the heater inlet tube.

When adding or topping of the engine coolant:

1. Measure the coolant concentration in the vehicle using the Coolant/Battery Refractometer 300-ROB75240 or equivalent.
2. Determine the concentration desired based on the vehicle duty cycle of extreme hot or cold operating conditions.
3. Add/top off or adjust coolant as follows:
 - For concentrations measured 48/52 to 50/50 (equates to a freeze point between -31°C [-30°F] and -34°C [-37°F]), use Motorcraft® Orange Antifreeze/Coolant Prediluted to maintain a coolant concentration in this same range.
 - For all other concentrations, use Motorcraft® Orange Antifreeze/Coolant Concentrated and/or distilled water to get to the desired concentration.
 - When refilling the engine coolant after a flush procedure, use a mixture of Motorcraft® Orange Antifreeze/Coolant Concentrated and distilled water to get to the desired concentration.
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3. Connect the heater inlet hose to the heater inlet tube and position the clamp.

All vehicles

4. Fill the radiator through the radiator fill neck until the coolant level is at the bottom of the filler neck.
When adding or topping of the engine coolant:
 1. Measure the coolant concentration in the vehicle using the Coolant/Battery Refractometer 300-ROB75240 or equivalent.
 2. Determine the concentration desired based on the vehicle duty cycle of extreme hot or cold operating conditions.
 3. Add/top off or adjust coolant as follows:
 - For concentrations measured 48/52 to 50/50 (equates to a freeze point between -31°C [-30°F] and -34°C [-37°F]), use Motorcraft® Orange Antifreeze/Coolant Prediluted to maintain a coolant concentration in this same range.
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5. Install the radiator cap.
6. Fill the coolant expansion tank until the coolant reaches the COLD FILL mark.
7. Select the maximum heater temperature and blower motor speed settings. Position the control to discharge air at A/C vents in instrument panel.
8. **NOTICE: If the air discharge remains cool and the engine coolant temperature gauge does not move, the engine coolant level is low in the engine and must be filled. Stop the engine, allow it to cool and fill the cooling system. Failure to follow this instruction may result in damage to the engine.**

Run the engine at idle until it reaches normal operating temperature.

9. Turn the engine off and allow it to cool.

10. Add the correct coolant mixture to the coolant expansion tank until the coolant level is at the COLD FILL mark.

11. Start the engine and allow it to idle until it reaches normal operating temperature. While the engine is idling, feel for hot air from the A/C vents.

- Hot air should discharge from the A/C vents. The engine coolant temperature gauge should maintain a stabilized reading in the middle of the NORMAL range and the upper radiator hose should feel hot to the touch.

12. Check the engine coolant level in the coolant expansion tank and fill it as necessary.

13. Repeat the previous 5 steps as necessary.